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CONTROLLING ELECTRICAL DEVICES
Using Motion Sensor

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Automating household or office systems can be achieved using an Arduino Nano, a passive infrared (PIR) motion sensor, and a relay. The primary aim is to create a control system that activates or deactivates electrical devices in response to human movement, enhancing convenience and energy efficiency.

The core component, the Arduino Nano, is a compact, versatile microcontroller board that processes signals from the PIR motion sensor. The PIR sensor detects changes in infrared radiation caused by human presence or movement within its range. When movement is detected, the sensor sends a signal to the Arduino Nano, which processes the input and triggers the relay. Fig. 1 displays the prototype.

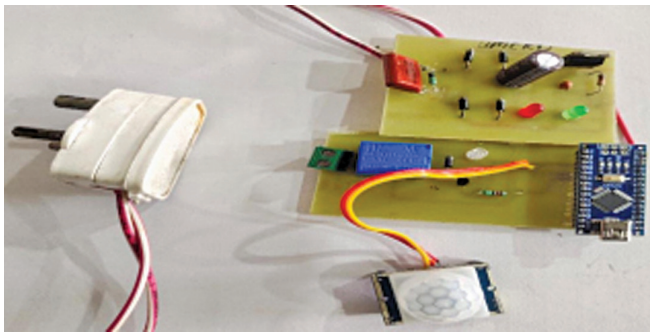


Fig. 1: Authors' prototype

The relay functions as an intermediary switch, controlling connected devices based on commands from the Arduino. This configuration enables the system to turn on lights when a person enters a room and switch them off after a set period of inactivity, thus optimising energy usage and user convenience.

Circuit and working

The circuit diagram of the automation device using a motion sensor is shown in Fig. 2. It consists of an Arduino Nano board (MOD1), a PIR sensor (S1), a 230V AC to 9V

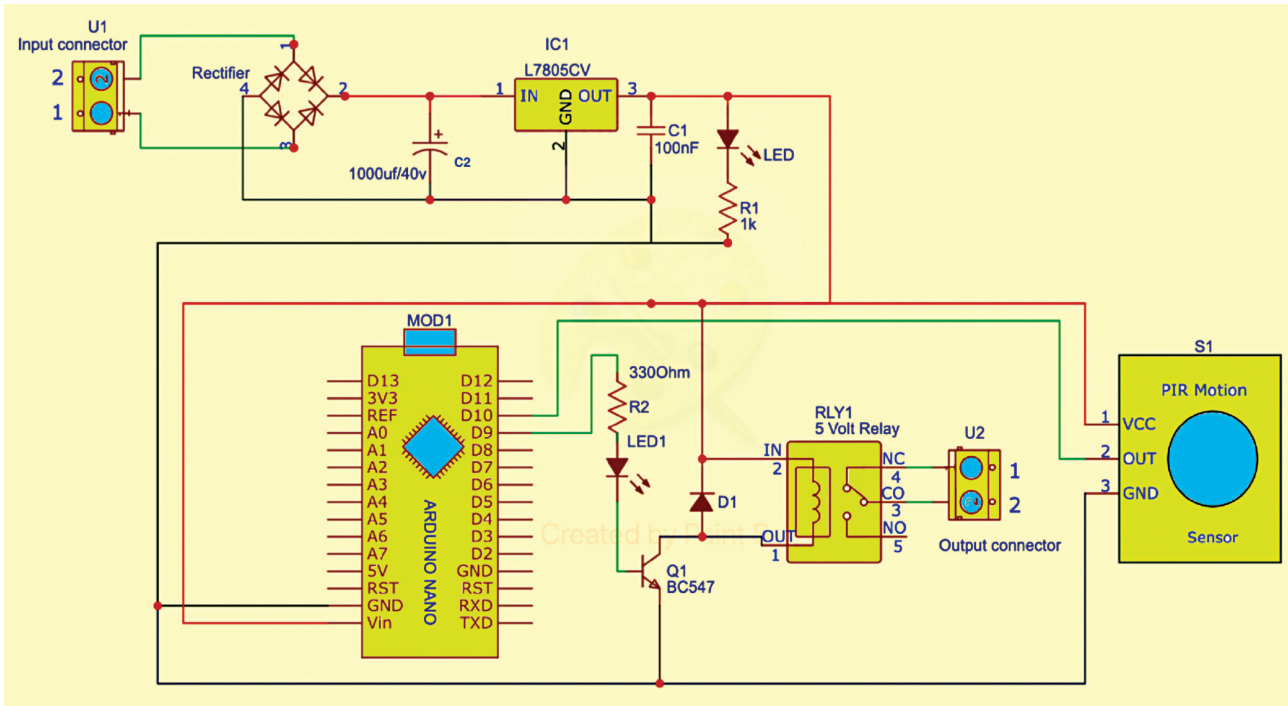


Fig. 2: Circuit diagram